

Functional traits influence patterns in vegetative and reproductive plant phenology – a multi-botanical garden study

Maria Sporbert^{1,2,3} , Desiree Jakubka^{1,2} , Solveig Franziska Bucher^{1,2} , Isabell Hensen^{1,3} ,
Martin Freiberg⁴ , Katja Heubach⁵ , Andreas König⁵, Birgit Nordt⁶ , Carolin Plos^{1,2,3} , Ilona Blinova⁷ ,
Aletta Bonn^{1,8,9} , Barbara Knickmann¹⁰, Tomáš Koubek¹¹ , Anja Linstädter¹² , Tereza Mašková^{11,13} ,
Richard B. Primack¹⁴ , Christoph Rosche^{1,3} , Manzoor A. Shah¹⁵ , Albert-Dieter Stevens⁶,
Katja Tielbörger¹⁶, Sabrina Träger^{1,3} , Christian Wirth^{1,4,17}  and Christine Römermann^{1,2} 

¹German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig 04103, Germany; ²Institute of Ecology and Evolution with Herbarium Haussknecht and Botanical Garden, Friedrich Schiller University Jena, Jena 07743, Germany; ³Institute of Biology/Geobotany and Botanical Garden, Martin Luther University Halle-Wittenberg, Halle (Saale) 06108, Germany; ⁴Institute of Biology, Leipzig University, Leipzig 04103, Germany; ⁵Palmengarten and Botanical Garden Frankfurt, Frankfurt am Main 60323, Germany; ⁶Botanic Garden Berlin, Freie Universität Berlin, Berlin 14195, Germany; ⁷Apatity, Murmansk Region 184209, Russia; ⁸Department of Ecosystem Services, Helmholtz-Centre for Environmental Research – UFZ, Leipzig 04318, Germany; ⁹Institute of Biodiversity, Friedrich Schiller University Jena, Jena 07743, Germany; ¹⁰Core Facility Botanical Garden, University Vienna, Vienna 1030, Austria; ¹¹Department of Botany, Faculty of Science, Charles University, Prague 12801, Czech Republic; ¹²Institute of Biochemistry and Biology, Department of Biodiversity Research/ Systematic Botany with Botanical Garden, University of Potsdam, Potsdam 14469, Germany; ¹³Ecology and Conservation Biology, Institute of Plant Sciences, University of Regensburg, Regensburg 93053, Germany; ¹⁴Biology Department, Boston University, Boston, MA 02215, USA; ¹⁵Department of Botany, University of Kashmir, Srinagar, Jammu & Kashmir 190006, India; ¹⁶Institute of Evolution and Ecology, University of Tübingen, Tübingen 72076, Germany; ¹⁷Max-Planck-Institute for Biogeochemistry, Jena 07745, Germany

Summary

Author for Correspondence:
Maria Sporbert
Email: maria.sporbert@gmail.com

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- Phenology has emerged as key indicator of the biological impacts of climate change, yet the role of functional traits constraining variation in herbaceous species' phenology has received little attention. Botanical gardens are ideal places in which to investigate large numbers of species growing under common climate conditions. We ask whether interspecific variation in plant phenology is influenced by differences in functional traits.
- We recorded onset, end, duration and intensity of initial growth, leafing out, leaf senescence, flowering and fruiting for 212 species across five botanical gardens in Germany. We measured functional traits, including plant height, absolute and specific leaf area, leaf dry matter content, leaf carbon and nitrogen content and seed mass and accounted for species' relatedness.
- Closely related species showed greater similarities in timing of phenological events than expected by chance, but species' traits had a high degree of explanatory power, pointing to paramount importance of species' life-history strategies. Taller plants showed later timing of initial growth, and flowered, fruited and underwent leaf senescence later. Large-leaved species had shorter flowering and fruiting durations.
- Taller, large-leaved species differ in their phenology and are more competitive than smaller, small-leaved species. We assume climate warming will change plant communities' competitive hierarchies with consequences for biodiversity.

Introduction

The timing of phenological events (such as leaf emergence, flowering, fruiting and leaf senescence) is crucial for species resource acquisition and reproductive success (e.g. plant–pollinator interaction, Nord & Lynch, 2009; Liu *et al.*, 2021). Phenology also has important implications for biotic interactions (e.g. herbivory) as well as for competitive hierarchies and ecosystem processes (Heberling *et al.*, 2019; Kudo & Cooper, 2019). Many studies

have demonstrated that plants are now leafing out earlier and flowering earlier in response to a warming climate (Blinova *et al.*, 2003; Root *et al.*, 2003; Wolkovich *et al.*, 2012; Bucher *et al.*, 2018; Menzel *et al.*, 2020; Rosbakh *et al.*, 2021). Plant functional traits (i.e. the morphological and physiological properties of plant species), affect species growth, reproduction and survival (Violle *et al.*, 2007). There is some evidence that differences in functional traits among species may also be associated with interspecific variation in plant phenology (Sun & Frelich, 2011;